CSC1042 Computer Network II

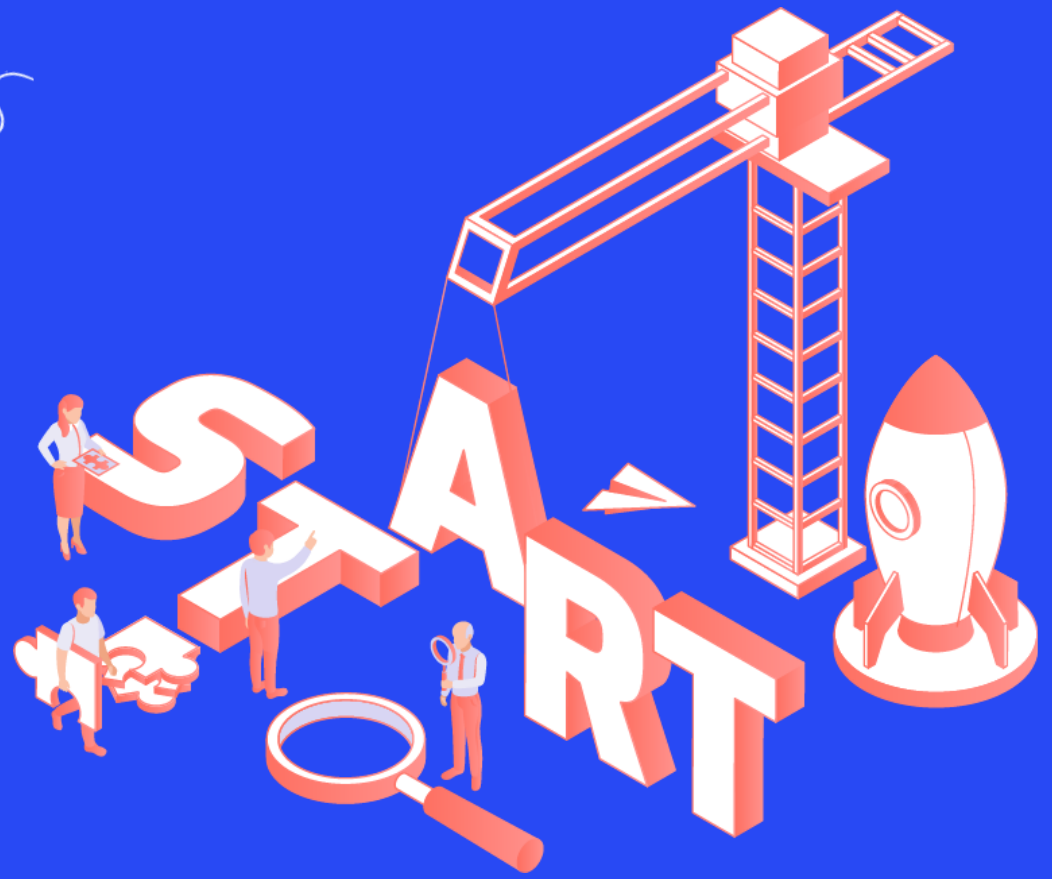
Address Resolution Protocol

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Learning outcomes

- Describe the service provided by ARP
- Explain how ARP works



Layered model

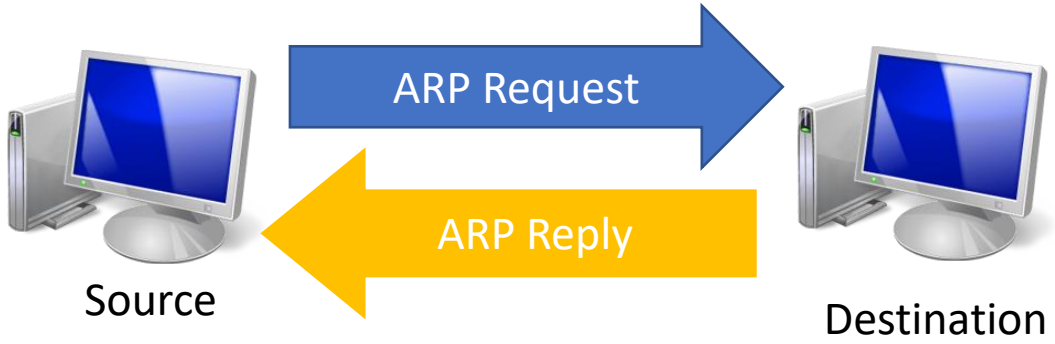
- We know that IP uses IP addresses to identify hosts and networks
- We also know that Ethernet used MAC addresses to identify host on a network
- In TCP/IP, a packet will be handed to Ethernet to be put in a frame before sending it through the medium
- Therefore, if a host wants to send a packet to a destination, it needs to know its IP and MAC addresses
- Assume the host knows the destination's IP address, how would it get its MAC address?

Address Resolution Protocol - ARP

- ARP maps IP addresses to MAC addresses
- It sits between the Network Layer and the Data Link Layer in the OSI model
- It resolves IP addresses to MAC addresses by sending request/reply messages
  - Example of request message: “Tell me who has IP address 192.168.10.5”
  - Example of reply message: “192.169.10.5 is at 00:0c:29:67:19:e4
- It keeps track of these mappings using an ARP table, called ARP cache

ARP – How it works?

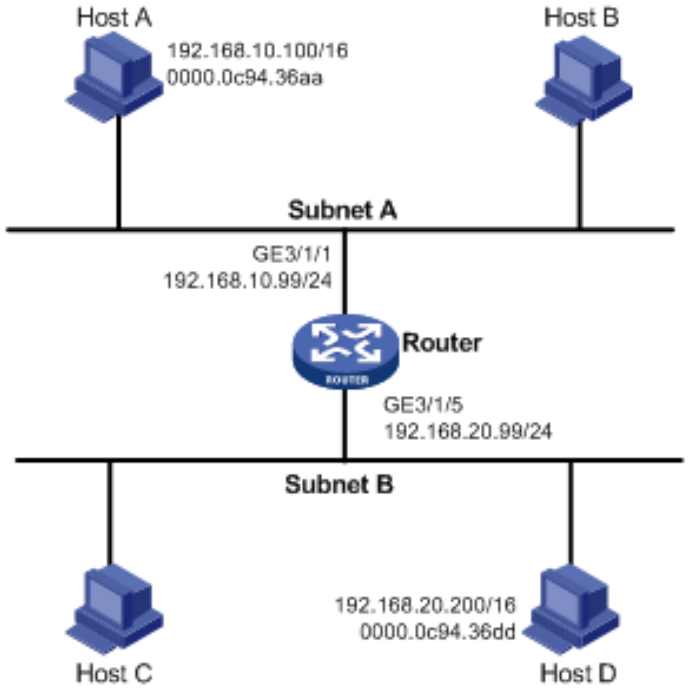
- When a source wants to send a packet to a destination on the same network which is not in its ARP cache, it starts the ARP process



  - The source broadcasts the ARP Request containing the IP address of the destination
  - Any host in the network receiving the ARP request will process it by comparing the included IP address against its own
    - If they match, the host (i.e., the destination) will send back an ARP reply to the source containing its MAC address.
    - Otherwise, the host drops the ARP Request

ARP – How it works?

- When Host A wants to send a packet to Host D which is not in its ARP cache, it starts the ARP process
  - Host A broadcasts the ARP Request containing the IP address of Host D
  - When received, the Router will know that the Host D is on another network, so it will send an ARP Reply with its own MAC address to Host A and will start the ARP process to get the MAC address of Host D



  - This is This is known as *Proxy ARP* since since routers do not forward broadcast ARP messages

THANK YOU